Back-ups and More with *rsync*

Learn about this powerful utility that almost all experts use in their day-to-day work, to perform tasks like back-ups, and much more.

The *rsync* utility can be used across platforms—on Linux, Mac OS X and Windows (with Cygwin, of course); and in combination with *cron* and SSH, it can easily be scripted. This makes it an essential utility in your tool-kit, even if you are not planning to use it for back-ups. Another advantage is that it is bundled with almost all major Linux distributions today. Its killer feature, really, is differential back-ups—*rsync,* with its unique algorithm. This allows you to transfer only the changes made in a file/directory tree, instead of re-transferring all data. This is very beneficial when synchronising large files or directory trees with gigabytes of data. *rsync* only transfers changed portions, and applies the changes to the file/directory tree copy on the other system, somewhat like the *patch* utility. It can even be used to synchronise files locally (on the same system), if you want to make back-ups on the local machine itself (say, to a different drive, like a USB drive). Overall, it is a simple, easy and efficient solution, where we don't even need to install any complicated back-up software.

Setting up and configuring *rsync*

Normally, *rsync* can directly be used by specifying source and destination directories, but it is usually set up in daemon mode (an '*rsync* server') at one end, so that it can receive requests for synchronisation. It can be set up in one-way or two-way synchronisation methods, as a standalone daemon configuration, or as an *inetd* configuration. The type of configuration used depends on the amount of traffic that your daemon is going to receive. For significant traffic throughout the day, it is better to have a standalone daemon; otherwise, the *inetd* configuration will do. Also, it is obvious that for two-way synchronisation, we have to run *rsync* in daemon mode at both ends. To configure *rsync* in daemon mode, modify the */etc/rsyncd.conf* file as follows:

```
motd file = /etc/rsyncd.motd
log file = /var/log/rsyncd.log
pid file = /var/run/rsyncd.pid
lock file = /var/run/rsync.lock
```